

Narrative Feature Extraction and Clustering from Humanitarian Journalism

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Overview

Natural Language Processing (NLP) offers a scalable approach to analysing discourse across large corpora. This project applies NLP methods to examine how narrative features in English humanitarian reporting differ across news outlets. This study aims to identify linguistic patterns that signal a Western-centric framing, particularly in public broadcasters.

RQ: To what extent can humanitarian reporting from different outlets be distinguished using narrative features alone, and which features drive the most separation as evidence of focalisation differences?

Methods

Dataset

- 1. Filtered BBC News Dataset (2005):** BBC News is a British national broadcaster. The original dataset was published in 2006 with 2,225 total articles. The filtered dataset includes 180 articles concerning foreign policy, global conflict, and international aid.
- 2. Self-curated The New Humanitarian (TNH) Dataset:** TNH is an independent nonprofit newsroom. I collected 180 articles from 2005 using their archive that include keywords on war, diplomacy, refugee, security, and international organisations.

Task Description

- 1. Text classification:** examines how distinguishable the articles are between the two sources.
 - A TF-IDF baseline model is trained using five-fold cross-validation
 - Two masked models, Named Entity Recognition (NER) masking and topic-based masking, test whether models rely on named entities or narrative structure.
- 2. Semantic role labelling (SRL):**
 - SRL operationalizes agency by identifying actors (ARG0) and patients (ARG1).
 - The SRL model analyses narrative tone and actor representation at the sentence level and aggregates patterns across each corpus.



Fig. 1: Text Classification Model-Performance Comparison
Even after NER masking, high performance persists → , structural features drive separation

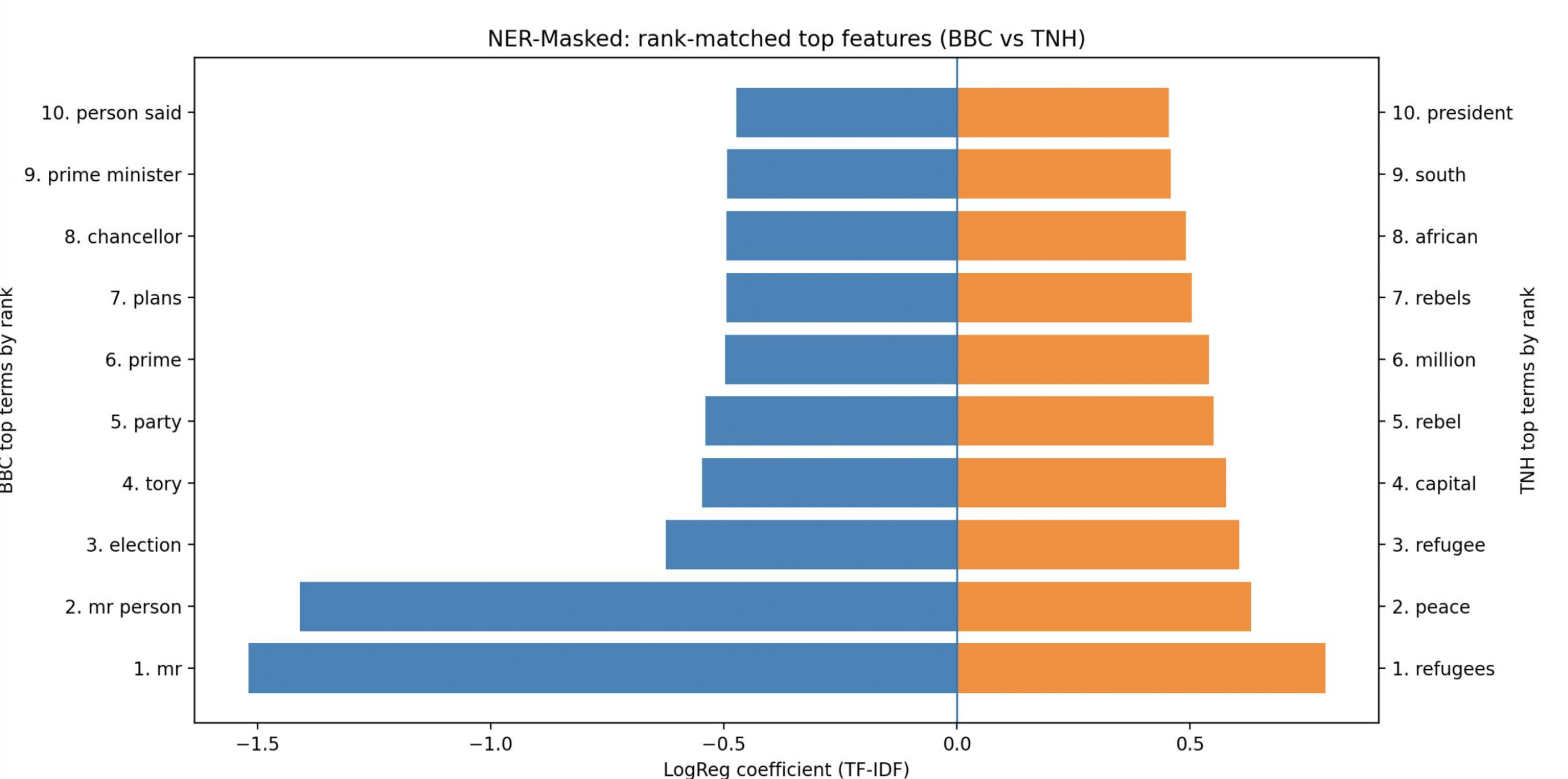


Fig. 2: Top Feature Extracted from NER-masked Model
BBC focused on named British government officials, and TNH focuses on groups such as “refugee”, “rebel,” and “African.”

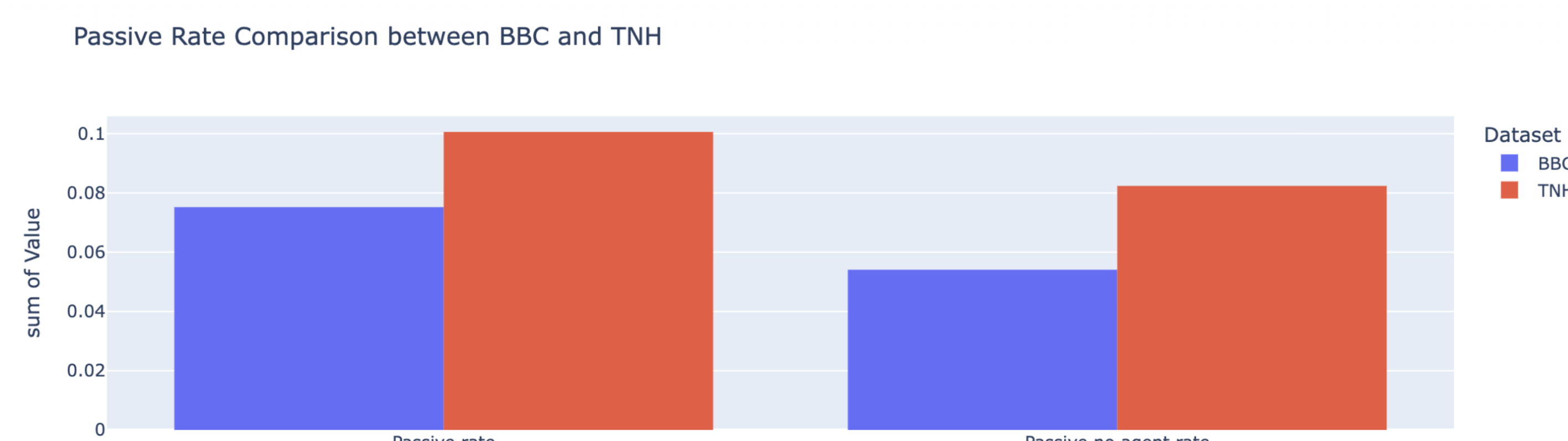


Fig. 3: Passive Rate in BBC and TNH dataset
TNH shows significantly higher agentless passive rates → TNH frames situations with suppressed agency

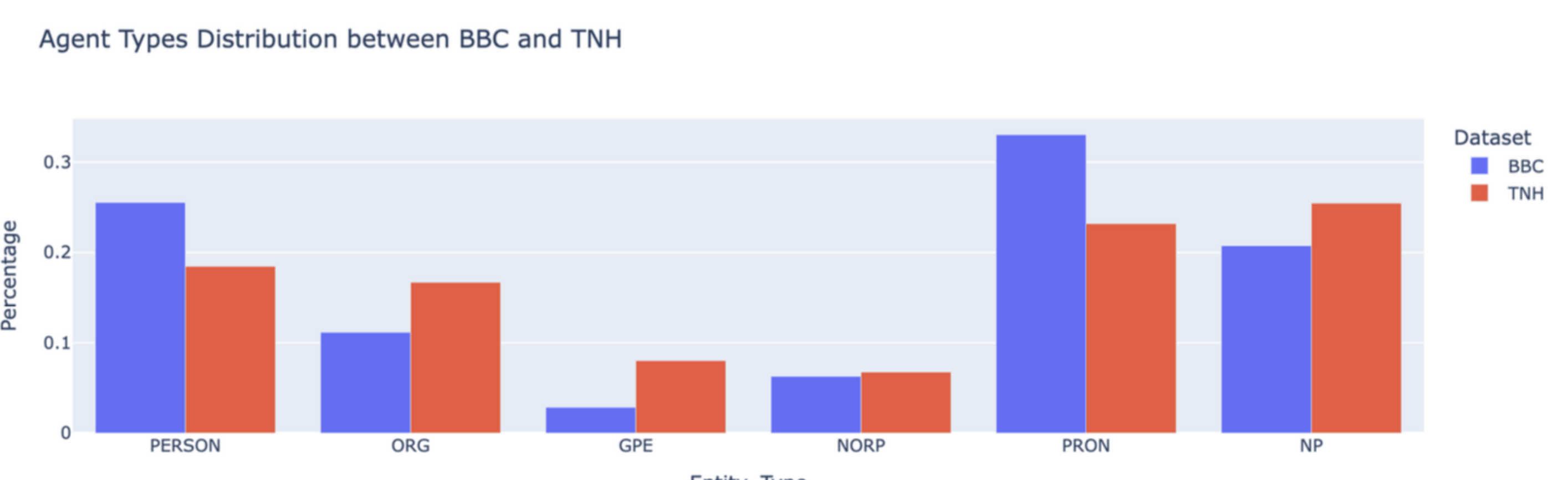


Fig. 4: Agent (ARG0) Type Distribution in BBC and TNH dataset: TNH prioritises complex agents while BBC focuses on a smaller number of figures.

Conclusion

Findings

- These results provide computational evidence for systematic differences in narrative focalisation across journalistic mandates.
- Classification and feature analyses reveal systematic narrative divergence between BBC News and The New Humanitarian, driven by TF-IDF features, voice, and semantic role structures.
- The New Humanitarian foregrounds affected populations as passive recipients with implicit agents.
- BBC News emphasizes elite political actors within a British geopolitical frame.

Future Work

- Improve causal validity through event-aligned corpora and stricter control of metadata artifacts
- Adopt neural models that better capture interactions among narrative features.
- Upgrade semantic role labelling with transformer-based approaches and introducing finer-grained, article-level metrics (e.g., agentless passive rates) would enable more precise analysis of narrative variation.

References

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